

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

Next item →

1.

What is the purpose given in the lessons this week for the iterated extended Kalman filter (IEKF)?

1 / 1 point

☒

To improve the accuracy of the EKF when the model's measurement equation is very nonlinear.

☐

To adaptively determine $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$.

☐

To decrease the computational complexity of the EKF.

☐

To decrease the memory requirements of the EKF.

☒ Correct

Yes, this is correct.
2.

What step or steps does the IEKF iterate? (Select all that apply.)

1 / 1 point

☐

It execute the prediction steps multiple times to improve the state prediction.

☐

It executes the output measurement step (of z_k) multiple times to make a better averaged measurement of z_k .

☒

It iterates the measurement update steps multiple times to improve the state estimate.

☒ Correct

Yes. That is not correct.

☐

It iterates the input measurement step (of u_k) multiple times to make a better averaged value of u_k .
3.

Execute the Coursera Lab Jupyter notebook for Lesson 3.3.2, which implements both the EKF and IEKF for a very challenging model.

1 / 1 point

Change the number of iterations from the default $N_I = 5$ to $N_I = 2$.

What is the root-mean-squared state-estimation error? Enter your answer with two digits to the right of the decimal point.

10.96

☒ Correct

Yes, that is correct.

4.

Execute the Coursera Lab Jupyter notebook for Lesson 3.3.2, which implements both the EKF and IEKF for a very challenging model.

1 / 1 point

Change the number of iterations from the default $N_I = 5$ to $N_I = 2$.

What is the percentage of time that the state estimate is outside of the estimation error bounds for the IEKF? Enter your answer with one digit to the right of the decimal point.

28.5

☒ Correct

Yes, that is correct.

5.

What is the purpose given in the lessons this week for the cubature Kalman filter (CKF)?

1 / 1 point

☐

To give better state estimates for systems having odd-order polynomial state equations.

☒

To simplify some aspects of the SPKF when process and sensor noises are additive.

☐

To adaptively determine $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$.

☐

To decrease the computational complexity of the SPKF.

☒ Correct

Yes, this is correct.

6.

What are some characteristics of the cubature Kalman filter (CKF)?

1 / 1 point

☐

It produces more accurate state estimates than other kinds of SPKF.

☒

It uses one fewer sigma point than the general SPKF.

☒ Correct

Yes, it deletes the sigma point for $i = 0$.

☒

It has no user-specified tuning parameters. That is, γ and $\alpha_i^{(m)}$ and $\alpha_i^{(c)}$ all have a pre-determined form.

☒ Correct

Yes, this is correct.

☒

The tuning parameters $\alpha_1^{(m)}$ and $\alpha_2^{(c)}$ are all non-negative, which makes the filter more robust to roundoff errors.

☒ Correct

Yes, this is true. It helps to preserve the positive-definiteness of the covariance calculations.

7.

Execute the Coursera Lab Jupyter notebook for Lesson 3.3.4, which implements both the EKF and CKF.

1 / 1 point

What is the rms error for the CKF for the **velocity** state? Enter your answer with four digits to the right of the decimal point. Note that the code is originally configured to compute the rms error for all the states (position and velocity) so you will need to make minor modifications to compute the desired result.

0.0024

☒ Correct

Yes. That is correct.

8.

What is the purpose given in the lessons this week of the adaptive extended Kalman filter (AEKF)?

1 / 1 point

☐

To improve the accuracy of the EKF when the model's measurement equation is very nonlinear.

☒

To adaptively determine $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$.

☐

To decrease the memory requirements of the EKF.

☐

To decrease the computational complexity of the EKF.

☒ Correct

Yes, this is correct.

9.

Execute the Coursera Lab Jupyter notebook for Lesson 3.3.6, which implements the AEKF. Change the variables **NV** and **NW** both to 500. (If you later need to change the code back to the original configuration, note that the default values for **NV** and **NW** are both equal to 1000.)

1 point

Compare the final AEKF estimates of $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$ to the true values. (You can find the true values in the **model** variable.) Which of the following is true? For this question, consider an estimate to be "good" if it is within 10% of the true value.

☐

The AEKF estimates of both $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$ are good.

☐

The AEKF estimate of $\Sigma_{\hat{w}}$ is good but the AEKF estimate of $\Sigma_{\hat{v}}$ is not good.

☒

The AEKF estimate of $\Sigma_{\hat{w}}$ is not good but the AEKF estimate of $\Sigma_{\hat{v}}$ is good.

☐

The AEKF estimates of both $\Sigma_{\hat{w}}$ and $\Sigma_{\hat{v}}$ are not good.

☒ Incorrect

No, this is not correct.

10.

For the same Coursera Lab Jupyter notebook, write down the RMS state estimation error and percentage of time that the error is outside of bounds when the variables **NV** and **NW** are both equal to 500.

1 / 1 point

Then, change the variables **NV** and **NW** both back to their default values of 1000. Rerun the code. Which of the following is true?

☐

Both the RMS state-estimation error and the percentage of time that the error is outside of bounds are better when the buffer sizes are 500 than when the buffer sizes are 1000.

☐

The RMS state estimation error is better when the buffer sizes are 500 than when the buffer sizes are 1000 and the percentage of time that the error is outside of bounds is worse when the buffer sizes are 500 than when the buffer sizes are 1000.

☒

Both the RMS state-estimation error and the percentage of time that the error is outside of bounds are worse when the buffer sizes are 500 than when the buffer sizes are 1000.

☐

The RMS state estimation error is worse when the buffer sizes are 500 than when the buffer sizes are 1000 and the percentage of time that the error is outside of bounds is better when the buffer sizes are 500 than when the buffer sizes are 1000.

☒ Correct

Yes, this is correct.